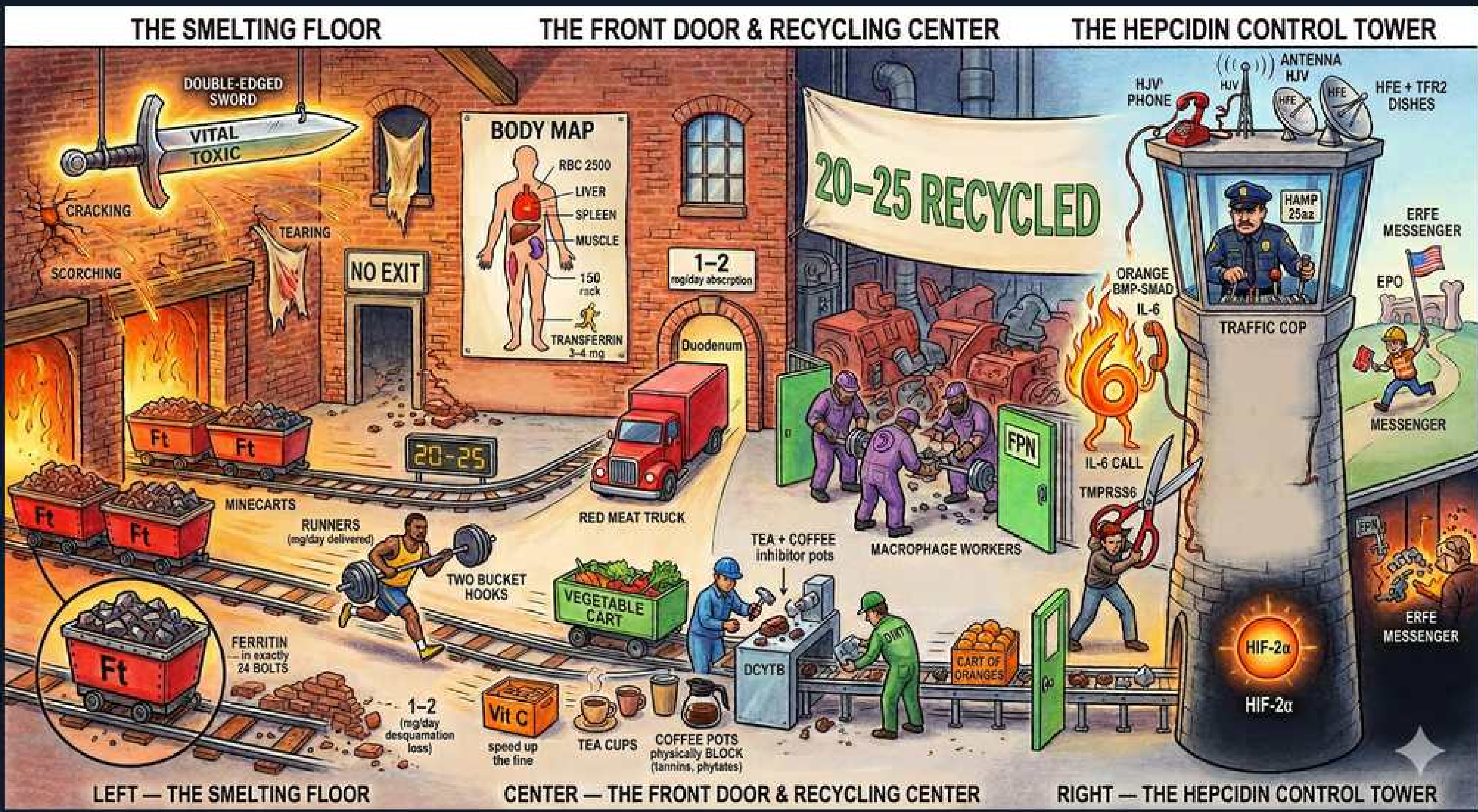
Iron / Microcytic — iron metabolism & hepcidin control



1. Smelting floor / absorption

WHAT YOU SEE

The left furnace floor with minecarts hauling ore into the smelter — the intake floor IS the DUODENUM, where dietary iron is absorbed (DCYTB reduces Fe^{3+}, DMT1 imports it).

WHAT IT ENCODES

Dietary iron is absorbed in the DUODENUM (and proximal jejunum). Ferrous Fe^{2+} is absorbed via DMT1 on the enterocyte apical membrane; ferric Fe^{3+} must first be reduced by duodenal cytochrome B (DCYTB). Heme iron (from meat) enters separately and is absorbed more efficiently than non-heme iron.

BOARD TRAP

WRONG: thinking iron is absorbed in the stomach or ileum. The duodenum is the site — which is exactly why celiac disease (duodenal villous atrophy) and gastric bypass (which bypasses the duodenum) cause iron deficiency. B12 is the one absorbed in the terminal ileum; do not swap them.

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2. Ferritin minecarts (stores)

WHAT YOU SEE

Carts labeled 'Ft', 'FERRITIN 4 BOLTS'

WHAT IT ENCODES

Ferritin is the intracellular iron STORAGE protein (a hollow shell holding up to ~4500 Fe atoms). Serum ferritin correlates with total body iron stores: roughly 1 ng/mL ferritin ≈ 8–10 mg stored iron. Low ferritin = depleted stores; ferritin <30 is specific for iron deficiency.

BOARD TRAP

WRONG: treating ferritin as a transport protein. Ferritin STORES iron inside cells; TRANSFERRIN transports it in plasma. Also remember ferritin is an acute-phase reactant — it rises in inflammation/malignancy/liver disease and can mask coexisting iron deficiency, so a 'normal' ferritin in a sick patient does not rule out IDA.

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3. 1–2 mg daily intake

WHAT YOU SEE

'1–2 mg daily' sign at bottom left

WHAT IT ENCODES

Only ~1–2 mg of iron is absorbed per day and ~1–2 mg is lost per day (sloughed enterocytes, skin, minor bleeding) — there is NO regulated excretion pathway. The body controls iron balance entirely at the level of ABSORPTION, which is why absorption is so tightly regulated by hepcidin.

BOARD TRAP

WRONG: assuming the body excretes excess iron. There is no active excretion route — this is precisely why transfusion-dependent patients and hereditary hemochromatosis develop iron OVERLOAD, and why overload is treated with phlebotomy or chelators, not by 'flushing it out.'

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4. Body map distribution

WHAT YOU SEE

BODY MAP poster: RBC 2500, liver, spleen, muscle

WHAT IT ENCODES

Total body iron ~3–4 g. Most is in hemoglobin within RBCs (~2500 mg / ~2/3 of total); the rest is in storage (ferritin/hemosiderin in liver and spleen macrophages ~1000 mg), muscle myoglobin (~300 mg), and a tiny labile plasma transferrin pool (~3 mg).

BOARD TRAP

WRONG: thinking the plasma/transferrin pool is large. Only ~3 mg of iron is circulating on transferrin at any moment — the bulk is locked in hemoglobin. This is why daily turnover depends overwhelmingly on macrophage RECYCLING of old RBCs, not on the small transit pool or diet.

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5. Transferrin truck

WHAT YOU SEE

Red 'RED MEAT' truck, 'TRANSFERRIN 3–4 mg', duodenum door

WHAT IT ENCODES

Transferrin is the plasma iron TRANSPORT protein; each molecule carries up to 2 Fe^{3+} ions and delivers iron to cells via transferrin receptor 1 (TfR1). TIBC (total iron-binding capacity) measures available transferrin: TIBC is HIGH in iron deficiency (more empty trucks) and LOW in anemia of chronic disease.

BOARD TRAP

WRONG: confusing transferrin with ferritin's role — transferrin does NOT store iron, it transports. Also: TIBC moves OPPOSITELY to ferritin/iron in IDA (iron low, TIBC high). A high TIBC with low TSAT points to true deficiency, whereas low TIBC suggests ACD.

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6. Vitamin C / absorption aid

WHAT YOU SEE

'Vit C' bottle, 'speed up the Fe', coffee/tea pots inhibit

WHAT IT ENCODES

Vitamin C (ascorbate) and gastric acid ENHANCE absorption by reducing $Fe^{3+} \rightarrow Fe^{2+}$. Inhibitors: tea/coffee (tannins/polyphenols), phytates (bran, legumes), calcium/dairy, and acid-suppressing drugs (PPIs, H2 blockers, antacids). Take oral iron on an empty stomach with vitamin C or orange juice.

BOARD TRAP

WRONG: telling a patient to take iron with food, milk, or their morning coffee/tea to 'avoid stomach upset.' These all blunt absorption; chronic PPI use is a frequently missed cause of refractory oral-iron failure. Best is empty stomach + vitamin C, spacing from antacids/coffee.

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7. 20–25 recycled banner

WHAT YOU SEE

Big banner '20–25 RECYCLED'

WHAT IT ENCODES

Macrophages of the reticuloendothelial system (spleen, liver) phagocytose senescent RBCs and recycle ~20–25 mg iron/day — roughly 10–20x more than the 1–2 mg from diet. This internal recycling is the dominant supply for daily erythropoiesis.

BOARD TRAP

WRONG: assuming most daily iron comes from the diet. Recycling dwarfs intake (~20–25 mg vs ~1–2 mg). This is why blocking macrophage iron release (high hepcidin in inflammation) rapidly starves erythropoiesis and causes anemia of chronic disease despite full stores.

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8. Macrophage workers / FPN

WHAT YOU SEE

Purple 'macrophage workers', 'FPN' exporter

WHAT IT ENCODES

FERROPORTIN (FPN) is the ONLY cellular iron EXPORTER — it moves iron out of enterocytes (into blood) and out of macrophages (recycled iron). Hepcidin is its off-switch: hepcidin binds ferroportin, causing its internalization and degradation, so high hepcidin traps iron inside cells.

BOARD TRAP

WRONG: thinking iron leaves cells passively or through DMT1. DMT1 imports iron INTO cells; ferroportin is the sole EXPORTER. The hepcidin–ferroportin axis is the single regulated step in human iron homeostasis — knock it out (low hepcidin) and you get hemochromatosis-like overload.

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9. Hepcidin control tower (HAMP)

WHAT YOU SEE

RIGHT tower traffic cop 'HAMP 25aa', antenna

WHAT IT ENCODES

Hepcidin (gene HAMP, a 25-amino-acid peptide) is the MASTER iron-regulatory hormone, made by the liver. It LOWERS serum iron by degrading ferroportin → less duodenal absorption and less macrophage release. Mnemonic: Hepcidin = 'Hides iron' (high hepcidin hides iron inside cells).

BOARD TRAP

WRONG: thinking hepcidin RAISES serum iron. Hepcidin LOWERS circulating iron. Hereditary hemochromatosis is fundamentally a hepcidin-DEFICIENCY state (HFE mutation → inappropriately low hepcidin → unchecked ferroportin → iron overload), the mirror image of inflammatory anemia.

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10. IL-6 raises hepcidin

WHAT YOU SEE

'IL-6 CALL', flames, signal up the tower

WHAT IT ENCODES

Inflammation drives IL-6 → JAK/STAT3 in hepatocytes → INCREASED hepcidin. Iron overload also raises hepcidin (negative feedback). High hepcidin degrades ferroportin → iron trapped in macrophages/enterocytes → the picture of anemia of chronic (inflammatory) disease: low serum iron, low TIBC, HIGH ferritin.

BOARD TRAP

WRONG: in a chronically inflamed patient, calling a low serum iron 'iron deficiency' and giving oral iron. High hepcidin blocks duodenal absorption, so oral iron WON'T work in inflammatory anemia — IV iron (which bypasses absorption) is needed if true coexisting deficiency is confirmed.

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11. HIF-2α lowers hepcidin

WHAT YOU SEE

Glowing 'HIF-2α' / 'HIF-2x' at tower base

WHAT IT ENCODES

Hypoxia, true iron deficiency, and increased erythropoietic drive LOWER hepcidin → more duodenal absorption and macrophage release. The bone-marrow signal is ERYTHROFERRONE (made by erythroblasts under EPO) which suppresses hepcidin; HIF-2α upregulates the duodenal absorption machinery (DMT1, DCYTB, ferroportin).

BOARD TRAP

WRONG: thinking hepcidin RISES in iron deficiency. In true iron deficiency hepcidin is appropriately LOW to maximize absorption. A high hepcidin in an anemic patient signals inflammation/ACD, not iron deficiency — the two are opposite hepcidin states.

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12. DCYTB / DMT1 uptake

WHAT YOU SEE

'DCYTB' worker at the recycling center

WHAT IT ENCODES

Duodenal cytochrome B (DCYTB) reduces dietary $Fe^{3+} \rightarrow Fe^{2+}$ at the enterocyte brush border, then DMT1 (divalent metal transporter 1) imports Fe^{2+} into the enterocyte. Inside, iron is either stored as ferritin (and shed) or exported to blood via ferroportin depending on hepcidin tone.

BOARD TRAP

WRONG: thinking ferric (Fe^{3+}) iron is absorbed directly. Non-heme iron must be REDUCED to Fe^{2+} (by DCYTB / acid / vitamin C) before DMT1 can take it up — which is why gastric acid and vitamin C aid absorption and PPIs/antacids impair it.
